Def. Let X be a Topological Space. A tuple of open sets $\{U, V\}$ is called Separation of X iff

$$U \neq \emptyset, \quad V \neq \emptyset, \quad U \cap V = \emptyset, \quad U \cup V = X.$$

If the separation exists, then X is called disconnected.

Thm. If X is Connected and $f: X \to Y$ is Continuous, then $f[X]$ is Connected subspace.

Proof. Suppose that $f[X]$ is disconnected. Let $\{U, V\} \subseteq \mathcal{T}_Y$ be a separation of $f[X]$.

That is, $U \neq \emptyset$, $V \neq \emptyset$, $U \cap V = \emptyset$, $U \cup V = f[X]$.

Then, $f^{-1}[U] \neq \emptyset$, $f^{-1}[V] \neq \emptyset$, $f^{-1}[U \cap V] = f^{-1}[U] \cap f^{-1}[V] = \emptyset$, $f^{-1}[U \cup V] = f^{-1}[U] \cup f^{-1}[V] = X$.

Since f is continuous, $\{f^{-1}[U], f^{-1}[V]\}$ is a separation for X. Contradiction.

Thm. If $A \subseteq X$ is Connected and $A \subseteq B \subseteq \overline{A}$, then B is Connected.

Proof. Suppose that B is disconnected. Then, there are opens $U, V \subseteq X$ such that

$$U \cap B \neq \emptyset, \quad V \cap B \neq \emptyset, \quad U \cap V \cap B = \emptyset, \quad U \cup V \supseteq B.$$

And, since $B \subseteq \overline{A}$, $U \cap \overline{A} \neq \emptyset$ and $V \cap \overline{A} \neq \emptyset$.

If $U \cap A = \emptyset$, then $A \subseteq U^c$, so $\overline{A} \subseteq U^c$. But this implies $\overline{A} \cap U^c = \emptyset$, contradiction.

Thus $U \cap A \neq \emptyset$, and similarly $V \cap A \neq \emptyset$. Moreover,

$$\emptyset = U \cap V \cap B \supseteq U \cap V \cap A, \quad \text{and} \quad U \cup V \supseteq B \supseteq A, \quad \text{so}$$

$\{U, V\}$ is also separation of A, but it contradicts.

Thm. If subspaces $A_\alpha \subseteq X$ $(\alpha \in \Lambda)$ are connected and $\bigcap_{\alpha\in\Lambda} A_\alpha \neq \phi$, then $\bigcup_{\alpha\in\Lambda} A_\alpha$ is connected.

proof. Denote $A = \bigcup A_\alpha$. Suppose that A is disconnected by the separation $\{U, V\}$.
That is, U, V are open s.t. $U \neq \phi$, $V \neq \phi$, $U \cap V = \phi$, $U \cup V = A$.
Let $a \in \bigcap_{\alpha\in\Lambda} A_\alpha$ be given. Then, for all $\alpha \in \Lambda$, $a \in A_\alpha$, thus $a \in A$.

If $a \in U$, put for each $\alpha \in \Lambda$, $U_\alpha = U \cap A_\alpha$ and $V_\alpha = V \cap A_\alpha$. Then,
$a \in U \cap A_\alpha \neq \phi$, $U_\alpha \cap V_\alpha = U \cap V \cap A_\alpha = \phi$, $U_\alpha \cup V_\alpha = (U \cup V) \cap A_\alpha = A_\alpha$.
Since A_α is connected, $V_\alpha = V \cap A_\alpha$ must be empty. But,

$$V = V \cap A = V \cap \left(\bigcup_{\alpha\in\Lambda} A_\alpha\right) = \bigcup_{\alpha\in\Lambda} (V \cap A_\alpha) = \bigcup_{\alpha\in\Lambda} V_\alpha = \phi.$$

Contradiction. If $a \in V$, it can be proved similarly. ∎

Corollary. If $A_n \subseteq X$ $(n \in \mathbb{N})$ are connected and for all $n \in \mathbb{N}$,

$$\left(\bigcup_{i=1}^{n} A_i\right) \cap A_{n+1} \neq \phi$$

then $\bigcup_{n=1}^{\infty} A_n$ is connected.
proof. Take $B_n = A_1 \cup \cdots \cup A_n$. And using induction. (Use $\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} B_n$)

Thm. Product of Connected space is Connected.

Proof. It holds in arbitrary product, but in this, prove any finite case

Let X, Y be Connected Space. For any $x \in X$ and $y \in Y$,

$$X \times \{y\} \cong X \quad \text{and} \quad \{x\} \times Y \cong Y$$

are Connected. Moreover, $X \times \{y\} \cap \{x\} \times Y = \{(x,y)\} \neq \emptyset$. Therefore

$$(X \times \{y\}) \cup (\{x\} \times Y)$$

is Connected. Let $b \in Y$ be fixed. Then, for any $x \in X$,

$$T_x = (X \times \{b\}) \cup (\{x\} \times Y)$$

is Connected, and $T_x \supseteq X \times \{b\}$. Thus, $X \times \{b\} \subseteq \bigcap_{x \in X} T_x \neq \emptyset$.

Now, $X \times Y = \bigcup_{x \in X} T_x$ is Connected.

\+ Arbitrary Product Case

Let X_α $(\alpha \in I)$ be Connected Spaces, and put $X = \prod_{\alpha \in I} X_\alpha$.

Let $(b_\alpha)_{\alpha \in I} \in X$ be fixed. Then, for any finite subset $\{\alpha_1, \cdots, \alpha_n\} \subseteq I$,

$$X_{(\alpha_1, \cdots, \alpha_n)} = \{(x_\alpha)_{\alpha \in I} \in X \mid \alpha \notin \{\alpha_1, \cdots, \alpha_n\} \Rightarrow x_\alpha = b_\alpha\}.$$

Then, $X_{(\alpha_1, \cdots, \alpha_n)} \cong X_{\alpha_1} \times \cdots \times X_{\alpha_n}$, So it is Connected.

Moreover, for a Collection $\mathcal{F}$ of every finite subsets of I,

$$\bigcap_{\{\alpha_1, \cdots, \alpha_n\} \in \mathcal{F}} X_{(\alpha_1, \cdots, \alpha_n)}$$ Contains (b_α), So non-empty,

$$Y = \bigcup_{\{\alpha_1, \cdots, \alpha_n\} \in \mathcal{F}} X_{(\alpha_1, \cdots, \alpha_n)}$$ is Connected.

So, $\bar{Y}$ is also Connected, enough to show $X = \bar{Y}$.

Let $\prod_{\alpha \in I} U_\alpha \subseteq X$ be a basis element of X. By def, all but finitely many $U_\alpha = X_\alpha$.

So, for some finite $\{\alpha_1, \cdots, \alpha_n\} \subseteq I$ s.t. $U_\alpha = X_\alpha$ if $\alpha \notin \{\alpha_1, \cdots, \alpha_n\}$

then $\prod_{\alpha \in I} U_\alpha \cap Y \neq \emptyset$. So proved.

Def. Let X be a Topological space. Define a relation $\sim$ on X as:

$$x \sim y \iff \exists \text{ connected subspace } A \subseteq X \text{ s.t. } x \in A \text{ and } y \in A.$$

Then, it is equivalent relation; reflexivity and Symmetricity are clear, and transivity: $x \sim y$ by A and $y \sim z$ by B, then $x \sim z$ by $A \cup B$.

Define Connected Component of X as:

$$C_x = [x]_\sim$$

Lemma. Let X be a Tps. Let $x \in X$ be given.

1). C_x is the largest Connected Subset Containing x.

2). If $A \neq \emptyset$ is connected subset of X, then there is unique Connected Component $C \supseteq A$.

3). Connected Component is closed.

Proof of 1). Let $A \subseteq X$ be a Connected Subset containing x. Then,

$$y \in A \Rightarrow x \sim y \Rightarrow y \in C_x, \text{ so } A \subseteq C_x.$$

Moreover, C_x is Connected, because: given $z \in C_x$, there is a Connected Subset A_z s.t. $x, z \in A_z \subseteq C_x$. Now, since $\bigcap_{z \in C_x} A_z$ Contains x, so nonempty, thus

$$C_x = \bigcup_{z \in C_x} A_z \quad \text{is Connected.}$$

2) is trivial, Fixed $x \in A$, $a \in A \Rightarrow a \in C_x$. The uniqueness is given by the relativity

3). $C_x \subseteq \overline{C_x}$ trivially, and $\overline{C_x}$ is connected, so $\overline{C_x} \subseteq C_x$.

「Thm. The Homeomorphism Preserves the Cardinality of Connected Component.

Proof. Let X and Y be Homeomorphic Topological Space with Homeomorphism $f: X \to Y$.
Given $x \in X$, there is a unique Connected Component of x, $C_x \subseteq X$.
Since the Connectedness is preserved by Continuous map, $f[C_x]$ is Connected subset,
Containing $f(x)$. So, $f[C_x] \subseteq C_{f(x)}$. Similarly, $f^{-1}[C_{f(x)}] \subseteq C_{f^{-1}(f(x))} = C_x$.
Therefore, for any $x \in X$, $f[C_x] = C_{f(x)}$ Similarly, given $y \in Y$, $f^{-1}[C_y] = C_{f^{-1}(y)}$.

Now, define a map $\phi: \{C_x \mid x \in X\} \to \{C_y \mid y \in Y\} : C_x \mapsto C_{f(x)}$

Then, ϕ is one-to-one:

$$C_{f(x_1)} = C_{f(x_2)} \Leftrightarrow f[C_{x_1}] = f[C_{x_2}], \text{ Since } f \text{ is Homeo, } C_{x_1} = C_{x_2}$$

(Take f^{-1} to the both sides.) And, given C_y from the Codomain,
there is $x \in X$ s.t $y = f(x)$. So $\phi(C_x) = C_{f(x)} = C_y$. ☒

「Thm. If a Topological Space X has only finite Connected Component,
then every Components are Clopen.

Proof. Let $C_1, \cdots, C_n$ be all connected Components of X. Then for each i,

$$C_i = X \setminus \left[\bigcup_{j \neq i} C_j \right]$$

is open.

Def. Let X be a Topological Space.
We said X is locally Connected at $x \in X$ if;
For any open $N \subseteq X$ containing $x \in X$, there is Connected open V s.t $x \in V \subseteq N$.

X is called locally Connected, if it is locally connected at every Point.

Thm. X is locally Connected if and only if:
for any open $U \subseteq X$, every Connected Component of U is open.

Proof. Suppose that X is locally Connected Space. Given an open set $U \subseteq X$, let $C \subseteq U$ be a Connected Component of the subspace U Since X is l.c connected, given $x \in C$, there is a Connected open set V_x s.t $x \in V_x \subseteq U$.
Since C is Connected Component of x, $V_x \subseteq C$. So,

$$C = \bigcup_{x \in C} V_x \quad \text{is open set.}$$

Conversely, let $x \in N$ be given. Then, there is open Connected Component $C_x \subseteq U$ exists.

Above definition and theorem are same in the case of Path-Connected.

Thm. If X is locally path-connected, then the Connected component and path component are same.
Proof. Let $C \subseteq X$ be a Connected Component Given $x \in C$, let P_x be a path component of x.
Since P_x is path-Connected, so Connected, thus $P_x \subseteq C$.
To show $P_x = C$, Suppose that $P_x \subsetneq C$. Let Q be the Union of;

$$\{P_y \mid P_y \text{ is a path-component of } X \text{ s.t } P_y \neq P_x \text{ and } P_y \cap C \neq \phi\}.$$

Then, for each P_y is Contained in C, So that $C = P_x \cup Q$.
Since X is locally path connected, P_x and Q are open, so these become the separation for C, it is Contradiction.

Corollary. If X is locally path-connected, then
Path Connected $\Rightarrow$ Connected.

Thm. Quotient map preserves locally connected by image.

Proof. Let X be a locally connected and $p: X \to Y$ be a quotient map.
Let $C \subseteq Y$ be a connected component of an open set U of Y.
Then, clearly $p^{-1}[C] \subseteq p^{-1}[U]$. Given $x \in p^{-1}[C]$, there is a connected component $C_x \ni x$.
Then, $p(x) \in p[C_x]$ is connected, so $p[C_x] \subseteq C$, or $C_x \subseteq p^{-1}[C]$.
So,
$$p^{-1}[C] = \bigcup_{x \in p^{-1}[C]} C_x$$
and since C_x are components of open set $p^{-1}[U]$,
So these are open, since X is locally connected.
Now, $p^{-1}[C]$ is open, so is C open, since p is quotient map. ☒

Thm. Quotient map preserves locally path-connected by image.

「Summary」

$$
\begin{tikzcd}
\text{Connected} \arrow[r, bend right] & \text{Path-Connected} \arrow[l, Rightarrow] \\
\text{locally Connected} & \text{locally Path-Connected} \arrow[l, Rightarrow] \arrow[u]
\end{tikzcd}
$$

「Def. Let X be a Topological Space.
If for any $a, b \in X$, there is a Continuous map

$$P_a^b : [0,1] \to X \quad \text{s.t } P_a^b(0) = a \text{ and } P_a^b(1) = b$$

then X is called path-Connected.

「Thm. Path-Connected $\Rightarrow$ Connected.
Proof. Suppose that X is disconnected, but path-Connected.
Let $\{U, V\}$ be the separation for X. Given $a \in U$ and $b \in V$,
there exists a Path $P : [0,1] \to X$ s.t $P(0) = a$, $P(1) = b$.
Since P is Continuous and $[0,1]$ is Connected, $\operatorname{Im} P$ is Connected, but

$$U \cap \operatorname{Im} P \ni a, \quad V \cap \operatorname{Im} P \ni b, \quad U \cap V \cap \operatorname{Im} P = \emptyset, \quad U \cup V \supseteq \operatorname{Im} P$$

So, $\{U, V\}$ is the separation for $\operatorname{Im} P$, it is Contradiction

「Thm. Let X be a Space. If $A_\alpha \subseteq X$ $(\alpha \in \Lambda)$ are Path-Connected with $\bigcap_{\alpha \in \Lambda} A_\alpha \neq \emptyset$,
then $\bigcup_{\alpha \in \Lambda} A_\alpha$ is Path-Connected.

Proof.

Topologist's Sine Curve.

Def. In $\mathbb{R}^2$, let $L = \{0\} \times [0,1]$ and $S = \{(x,y) \mid x > 0,\ y = \sin\frac{1}{x}\}$.
Then, the subspace $L \cup S$ is called the Topologist's Sine Curve.
It has the following properties;

1). Connected.
Since $x \mapsto \sin\frac{1}{x}$ $(x>0)$ is Continuous, the graph S is Connected.
And, since $L \cup S \subseteq \overline{S}$, $(L \subseteq S'$, so $L \cup S \subseteq S')$,
$L \cup S$ is also Connected, by A connected $\Rightarrow$ B s.t. $A \subseteq B \subseteq \overline{A}$ is connected.

2. Not Path-Connected.
S is path-Connected, Since $(0, \infty)$ is path-Connected and $x \mapsto \sin\frac{1}{x}$ is continuous.
But, for $(0,0) \in L$ and $(1, \sin 1) \in S$, there is no path between $(0,0)$ and $(1, \sin 1)$
(This is also the Counter-example for S path. conn. but $\overline{S}$ is not.)
If there is a continuous map $p : [0,1] \to L \cup S : t \mapsto (x(t), \sin(\frac{1}{x(t)}))$ s.t. $p(0) = (0,0)$, $p(1) = (1, \sin 1)$,
the coordinates x and y are continuous. Now, define

$$t_n = \frac{2}{(2n+1)\cdot\pi}.$$

Then, $t_n \to 0$ as $n \to \infty$, but $y(t_n) = (-1)^n \not\to 0$, does not converges to zero.
So, Contradiction.